

Prathamesh Bhavsar

Dallas, TX | prathameshbhavsar19@gmail.com | +1(940)279-9009 | GitHub | Website | LinkedIn

Summary

AI/ML Engineer with about 4 years shipping production ML systems, the last 1.5 focused on LLM/RAG and Generative AI. Track record of taking models from prototype to monitored production in regulated, high-stakes domains such as energy and healthcare. Started in classical ML and data infrastructure, then carried that same focus on evaluation, reliability, and observability into LLM/RAG and agentic systems.

Skills

- **Programming & Databases** - Python, SQL, Spark, Snowflake, MongoDB, NumPy, Pandas
- **AI/ML & GenAI** - TensorFlow, PyTorch, LangChain, GPT-4, BioBERT, RAG, Vector Search, Prompt Engineering
- **Cloud & MLOps** - AWS, SageMaker, Bedrock, EC2, S3, Step Functions, MLflow, Airflow, Docker, CI/CD

Experience

Data and AI Co-Op, Tenet Healthcare - Dallas, TX

Aug 2025 – May 2026

- Contributed to a production **GenAI medical-coding assistant** built on **LangChain** and **GPT-4** through **Azure OpenAI**, working alongside clinical and engineering teams to keep **PHI de-identified under a signed BAA**; improved coding accuracy by **about 9 points** and reduced manual coding effort by **around 40%**.
- Built a **clinical-trial-matching prototype** using **BioBERT embeddings** and **graph-based retrieval** that increased patient-screening throughput and surfaced **roughly 30% more** eligible candidates for review.
- Helped operationalize a real-time **clinical decision-support tool** across **ICU settings**, implementing latency and drift dashboards plus stakeholder reporting to keep the system observable and compliant in production.
- Partnered with healthcare, data, and engineering stakeholders to align model behavior with compliance-sensitive workflows and clinical validation requirements before release.

Software Development Engineer (Machine Learning), EnelX - Mumbai, India

Jul 2021 – Aug 2024

- Developed and deployed scalable **predictive-maintenance ML services** on high-throughput sensor data, reducing unplanned equipment outages by **around 35%** through earlier fault detection.
- Contributed to an **energy-analytics platform** that turned enterprise telemetry into cost-optimization insights, helping cut average energy spend by **about 18%** across **300+ facilities**.
- Built distributed **data and ML pipelines** for streaming IoT and energy telemetry, adding monitoring and lineage that cut mean-time-to-root-cause for pipeline failures from **about 2 hours to under 40 minutes**, roughly 66% faster.
- Built optimization models for **EV-fleet operations** that improved vehicle uptime by **roughly 6 percentage points** across **800+ vehicles**.
- Led **data-quality analysis** across **30+ suppliers**, cutting duplicate-record overhead from **about 12% to 5%** and reclaiming close to **40 hours** of manual reconciliation monthly.

Education

University of Texas at Dallas, MS in Computer Engineering

Aug 2024 – May 2026

University of Mumbai, Bachelor of Engineering in Information Technology

Jul 2017 – Jun 2021

Projects

PitWall — Production RAG API over F1/FIA Regulations (FastAPI, Qdrant, GPT-4o, Langfuse)

[GitHub Link](#)

- Built and deployed a production **hybrid-RAG API** (dense, BM25, and reciprocal rank fusion retrieval with GPT-4o generation) over FIA regulation text, reaching **0.84 context precision** and **0.87 faithfulness** on RAGAS.
- Added a CI/CD pipeline with automated **RAGAS evaluation gates** and request-level observability (cost, latency, and token tracing via Langfuse); load-tested to **150 concurrent users** on Azure Container Apps.

AgentToyota — Smart Vehicle Financing AI Agent (FastAPI, React, LangChain)

[GitHub Link](#)

- Built a full-stack **LLM agent** using FastAPI, React, and LangChain that orchestrates tool calls and structured outputs to automate financing-decision steps, reducing manual decision time by **about 30%** while logging decision quality, not just speed.

StreamFraud: Real-Time Credit Card Fraud Detection (Kafka, Docker, Isolation Forest)

[GitHub Link](#)

- Built a containerized **real-time anomaly-detection pipeline** using Kafka and Docker for streaming transactions, evaluated on the imbalanced fraud class with precision, recall, and PR-AUC (**about 0.83 precision, 0.71 recall, and 0.78 PR-AUC**) rather than raw accuracy.